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Review

Towards sustainability and a circular Economy: ROMP for the goal of fully degradable and chemically recyclable polymers

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ABSTRACT

Over the past century the advancement of polymer chemistry has provided civilization with a plethora of plastic products that have significantly contributed to day-to-day life and the general well-being of humanity. However, the disposal of used plastic has led to an accumulation of plastic waste in the environment which can destroy fragile ecosystems. It is therefore imperative that viable alternatives to the presently used non–degradable polymers are developed. Two emerging approaches for addressing the contemporary plastic waste crisis are the development of degradable and chemically recyclable polymers. Among the various polymerization techniques, ring–opening metathesis polymerization (ROMP) remains a viable candidate for the design and preparation of such next-generation polymers. Within this review article an overview is presented regarding the history, mechanism and catalyst developments in ROMP, detailing significant developments over the previous decades which reassert the importance of ROMP in designing advanced polymers. Moreover, this article aims to explore the chemistry involved in ROMP by summarizing the state-of-the-art, as well as providing an overview of the fundamental challenges involved in synthesizing such next-generation polymers, including potential future directions.

1. Introduction

Innovations in polymeric materials, plastics in particular, have dramatically changed our everyday lives. Due to their low-cost of production, light weight, durability and environmental resistance, plastic has dominated the global market as the most frequently used material for the production of daily-life products.[1] The annual worldwide production of plastics exceeds 335 million tonnes, and is projected to reach triple this number (1.2 billion tonnes) by 2050.[2] Unfortunately, the exponential rise in plastic production, alongside their often single-use nature (for example, food packaging) and extremely slow rate of decomposition,[3] have significantly contributed to environmental pollution and climate change.[4] In a 2018 case study it was reported that out of the 250 million tons of plastic waste generated globally among surveyed countries, 20 % (50 million tons) were incinerated for energy recovery, 20 % (50 million tons) was recycled, 30 % (75 million tons) was disposed of in landfills, 25 % (62 million tons) were disposed improperly, and 5 % (13 million tons) were leaked into the environment.[5] Leakage of plastic waste into terrestrial and aquatic environments significantly affects biodiversity and the ecosystem. Furthermore, contamination of the food chain by microplastics is also a major concern for human health.[6] As a consequence, the proper management of plastic waste is one of key challenges presently facing humanity as it transitions into an environmentally conscious future.

A promising approach for reducing plastic waste is the development of degradable polymers. These polymers are typically derived from biorenewable monomers, for which the corresponding polymer can be decomposed into CO_2, CH_4, water and humic acid by enzymatic or hydrolytic activity, leading to an environmentally closed system.[7] Significant efforts have been dedicated to the chemistry and design of degradable polymers. An emerging technique, ring–opening polymerization (ROP), is a powerful strategy to incorporate cleavable linkages (e.g. esters, ethers, thioesters, etc.) into the polymer backbone.[8,9] Such linkages impart degradability to the polymers, with only a minor

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quantity actually being required to be introduced into the polymer backbone for complete polymer degradation, assuming homogeneous incorporation. Furthermore, degradable polymers with hydrolysable groups such as acetal, ether and ester groups in the polymer backbone have found numerous applications in the fields of biomedicine,[10] lithography,[11] energy storage, and electronics.[12].

Another emerging approach for reducing plastic waste is chemically recyclable polymers. Chemical recycling involves a process in which post-consumer polymer waste is depolymerized under controlled conditions to yield virgin monomer feedstock or other value added pre-monomers, which can then be purified and repolymerized.[13] It is envisioned that chemical recycling could transform the polymer industry from a linear economy to a genuine circular economy, enabling a closed-loop system for the use of polymer materials and thereby solve the end-of-use problem of modern plastics.[7] Recently, significant research interests have been focused on developing new chemically recyclable polymers, which may begin to play a prominent role in the field of sustainable chemistry.

As a prominent ROP method, ring-opening metathesis polymerization (ROMP) has emerged as one of the most widely applied polymerization techniques for synthesizing structurally and functionally diverse polymer architectures from cyclic monomers.[14–17] As a transition metal catalyzed method, ROMP allows for the synthesis of polymers with well-controlled molecular weights and dispersities.[18–21] Because of the wide utilization of carbocyclic olefin monomers, most ROMP–derived polymers lack backbone degradability. However, recent developments have revealed that degradable ROMP-derived polymers can also be synthesized by the proper design of the cyclic olefin monomer so that labile or degradable linkages can be incorporated into the polymer backbone.[22,23] These strategies have led to the development of a diverse array of polymer architectures including homopolymers, block and alternating copolymers which display full degradation. Mechanistic investigations have revealed that all steps within ROMP are principally reversible, and therefore exist as an equilibrium between polymerization and depolymerization.[24] Therefore ROMP is an ideal polymerization technique for the design of chemically recyclable polymers.

The development of degradable and chemically recyclable polymers via ROP techniques (including ROMP) is a broad research area with several recent developments. Accordingly, there have been several excellent review articles published[13,25,26] that partially overlap with some of the topics discussed herein. In consideration of the significance of ROMP towards addressing challenges related to the development of a sustainable and/or circular economy, we present this review article focusing on the importance of ROMP in synthesizing degradable and chemically recyclable polymers. The article is divided into multiple sections; within the first section a short history, detailed mechanism, and important catalyst developments for ROMP are outlined. Following this, the synthesis of ROMP-derived degradable and chemically recyclable polymers are reviewed in detail. Within these sections, a detailed outline of the monomers, catalysts, molecular weights, dispersities, and degradation or depolymerization conditions are provided to provide a thorough overview.

2. History of ROMP

In 1967, Calderon first defined the term 'olefin metathesis' after studying mechanisms involved in the ring-opening polymerization of cyclic olefins and the disproportionation reactions of acyclic olefins.[27] However, much of the pioneering work was carried out by researchers at DuPont into a then-unnamed metathesis reaction while investigating a new process for the synthesis of polypropylene.[28] Following these pioneering works, various researchers attempted to elucidate the mechanism for this metal–catalyzed reaction in order to explore its potential scope within synthetic chemistry. Following initial studies on this reaction by Calderon,[29,30] in 1971 Hérrison and Chauvin, were pioneers in proposing the now accepted mechanism for olefin metathesis, which proceeds through a metallocyclobutane intermediate, with later significant developments by R. H. Grubbs and others.[31–33].

3. Mechanism of ROMP

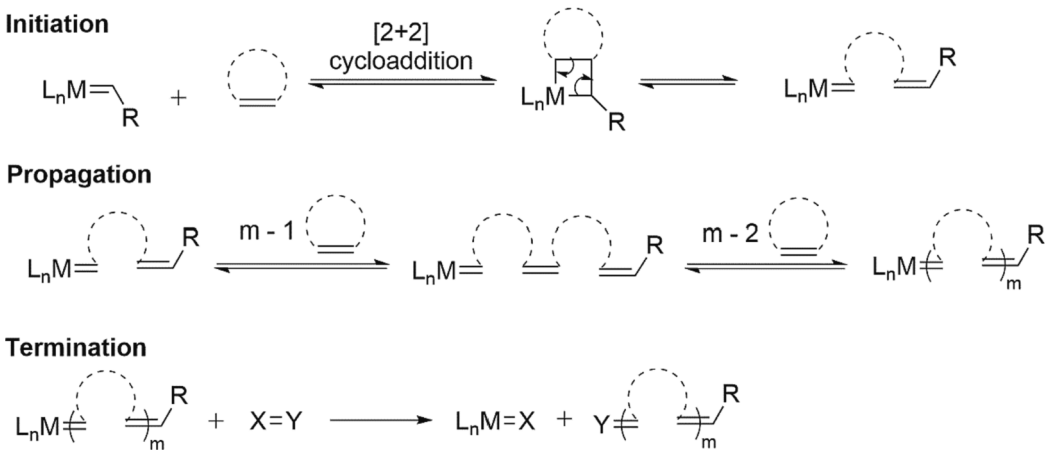
Ring-opening metathesis polymerization (ROMP) is a transition metal alkylidene-triggered process in which cyclic olefins undergo ring-opening and are concomitantly joined together to form a polymer.[34] Similar to Ziegler-Natta type polymerizations, ROMP is a chain-growth polymerization that belongs to the family of polyinsertions and therefore encompasses at least three discrete steps: initiation, propagation and termination (Scheme 1). Initiation begins with the coordination of a transition metal alkylidene complex to a cyclic olefin to form a metallocyclobutane intermediate, followed by [2 + 2] cycloaddition, and finally cycloreversion to generate a linear propagating species. In principle, all the steps within ROMP are reversible and therefore exist as an equilibrium between polymerization and depolymerization.[24] Recent developments in ROMP provide a versatile strategy for the construction of functional polymers with controlled molecular weights and architectures, including block copolymers.[24,35,36] In addition, ROMP displays living polymerization characteristics for many monomer types, particularly strained and bicyclic olefins.[37].

To obtain detailed insights on the mechanism of ROMP, the initiation step can be considered to operate via three plausible mechanisms; associative, dissociative and interchange mechanisms. In the 'associative' mechanism, the olefin monomer coordinates to the metal alkylidene catalyst prior to ligand displacement, while in the 'dissociative' mechanism the ligand is lost prior to olefin binding. In the 'interchange' mechanism the olefin binding and the loss of the ligand occur simultaneously through a common transition state.[38] The mode of initiation is anticipated to be dependent upon the catalytic species, and as a result the mechanism and rate of initiation can be tuned by proper design of the catalyst i.e. varying the stereoelectronic nature of the ligands around the metal center. Detailed insights on the modes of initiation can be found in an excellent article by Nolan and coworkers, which also includes recent developments on the mechanistic studies of ROMP.[39].

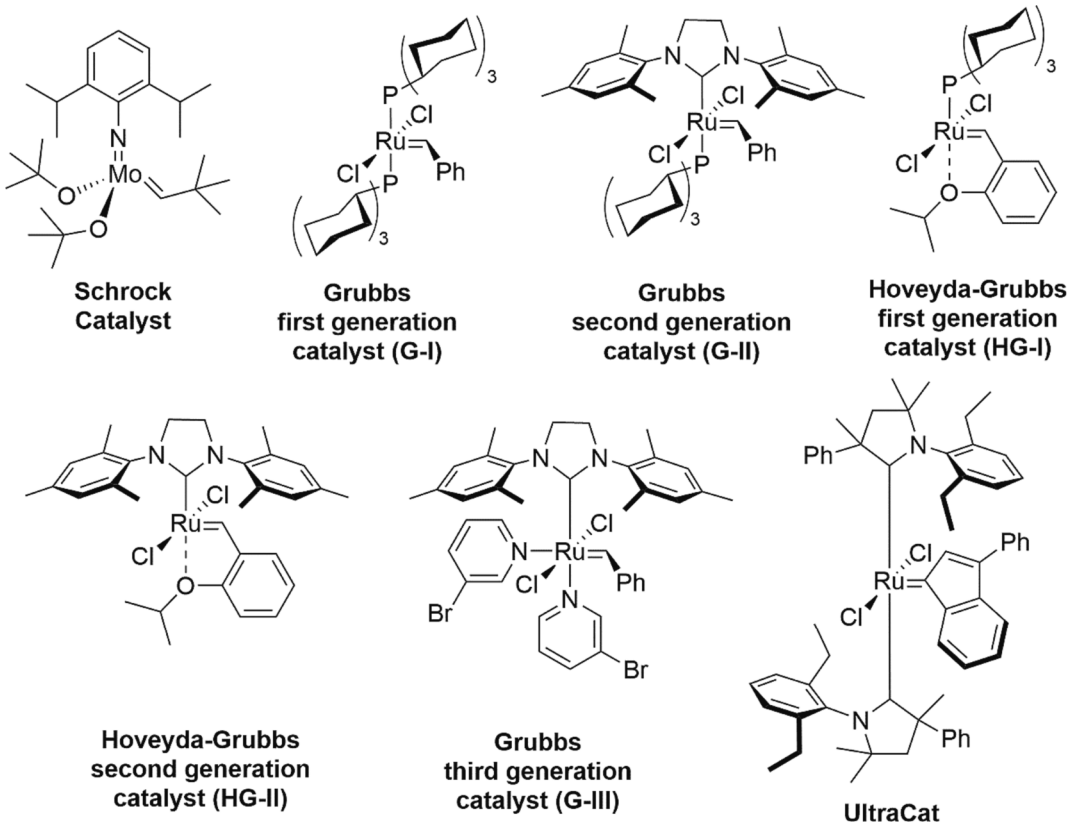
The propagation step involves the coordination of a 14-electron metal alkylidene complex (common for all Ru-based metathesis catalysts) with the cyclic olefin monomer to grow a polymer from the metal center, with the addition of monomer units via insertion at the ω-chain end. The propagation process continues until: a) the consumption of all monomer, b) the establishment of a thermodynamic equilibrium or, c) termination of the reaction by deactivation of the catalyst.[24] Termination in ROMP is a unique process when compared to other living polymerization techniques. Instead of chain-termination via intermolecular reaction between two chains, the termination is generally carried out using quenching reagents (e.g. vinyl ethers), which lead to the formation of non-reactive Fischer carbene species, [40] or by uncontrolled decomposition of the catalyst.[24,41–44].

4. Catalysts for ROMP

Intense interest in the catalytic ROMP process began to develop in the 1970s. The initial catalysts for ROMP were ill-defined and poorly understood, and were derived from early transition metals such as tungsten, molybdenum and rhenium, and were used combined with aluminum-based Lewis acid co-catalysts (e.g. EtAlCl). The mode of action for these catalytic systems towards the ROMP mechanism was also not well defined, and their reactivity scope was severely limited.[45] In the mid 1970s the first well-defined, homogeneous, single-component ROMP catalysts emerged based on organometallic complexes of tantalum and tungsten which facilitated milder reaction conditions along with a better mechanistic understanding.[46,47] Later, the subsequent discovery of well-characterized Mo- and Ru-complexes provided the first commercially available metathesis catalysts (Scheme 2). A



Scheme 1. General mechanism of ring-opening metathesis polymerization (ROMP).



Scheme 2. Structures of the common and commercially available metathesis catalysts.

detailed history of ROMP catalyst development has been already reviewed in depth[32,45] and thus we only intend to focus on commercially available ROMP catalysts based upon Mo and Ru (Scheme 2).

In the mid-1970s, Schrock pioneered the first isolation of a tantalum alkylidene complex to catalyze ROMP, which was subsequently followed by the introduction of tungsten–based alkylidene complexes.[46,48] The first commercially available ROMP catalyst was developed by the same group in 1990, and was marketed as 'Schrock's catalyst', a Mo-based alkylidene complex with a remarkably high catalytic activity for metathesis reactions, leading to its extensive utilization in ROMP during the subsequent decades.[49] In the late 1990s, Grubbs and coworkers were the first to report the Ru-based metathesis catalyst now commonly known as the 'Grubbs 1st generation catalyst' [**G-I**, $(PCy_3)_2(Cl)_2RuCHPh$] which became the first of several commercially available catalysts in this series.[50–53] The chief practical difference between these two classes of metathesis catalyst are that the Mo-based catalysts were somewhat more active but were also sensitive to air and moisture, alongside reacting with some common functional groups. In contrast, the Ru-based catalysts are generally bench stable and show greater functional group tolerance (i.e., reactions can be carried out in presence of functionalities such as aldehydes, ketones, esters, and amides). Additionally, Ru centers have a specific affinity for C=C bonds, advantageous for the polymerization of monomers with pendant functional groups that often cannot be polymerized using other catalysts. Later, Grubbs and coworkers focused their efforts on understanding the mechanism of **G-I** catalyzed metathesis reactions and noted that the initiation step in these transformations proceeded via the 'dissociative'

mechanism.[38] Thus, both the initiation and propagation steps require the dissociation of phosphine from the Ru center to generate a catalytically active 14-electron Ru–intermediate species. In an effort to accelerate the dissociative step, Grubbs and coworkers introduced a strong σ–donating *N*-heterocyclic carbene (NHC) ligand in place of one phosphine ligand, this pioneering work leading to the development of the Grubbs 2nd generation catalyst (**G–II**) with 100 times higher activity, which is now a widely used catalyst in metathesis reactions.[54,55] Being a strong σ-donor and very poor π-acceptor, NHC ligands form a strong bond with the metal centers which better stabilize the 14–electron active Ru-species and can be employed as an alternative to the widely used phosphine ligands. These contributions toward the development of olefin metathesis and catalyst discovery provided the grounds for the 2005 Nobel Prize in chemistry, jointly awarded to Chauvin, Grubbs and Schrock.

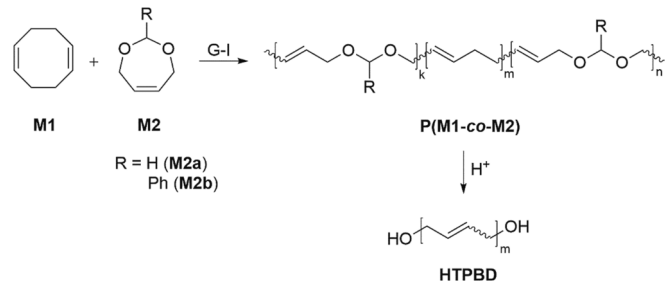
In 1999, Hoveyda and coworkers investigated the utilization of styrenyl ethers as bidentate benzylidene ligands and demonstrated that **G-I** forms stable Ru-chelates in the presence of styrenyl terminal olefins, and that the corresponding Ru-chelate complexes are still metathesis-active.[51] The introduction of such ligands dramatically increased the stability of the catalysts, allowing catalyst to be readily recovered and recycled after a metathesis reaction. The chelating oxygen of the bidentate benzylidene ligand replaces a phosphine ligand, which in the case of **G-II** catalyst gives a completely phosphine-free structure. As a consequence, two additional commercially available metathesis catalysts were developed; namely, Hoveyda-Grubbs 1st (**HG-I**) and Hoveyda-Grubbs 2nd generation catalyst (**HG-II**).[50,51,56] The enhanced activity of the **HG-II** catalysts in olefin metathesis was hypothesized to be due to the absence of phosphine ligands, which have been thought responsible for suppressing catalytic activity through competition with olefins for the open ligation sites on the Ru-based active catalytic species.

Problems in applying **G-II** catalyst to living ROMP involved its slow initiation rate with respect to that of monomer propagation. This was overcome in 2002 by developing a new Ru catalyst bearing labile pyridine ligands, now commercially known as Grubbs 3rd generation catalyst (**G-III**).[57] **G-III** catalyst can be readily synthesized from the commercially available **G-II** catalyst by its treatment with pyridine or a substituted pyridine.[54,58] Because of the labile pyridine ligands, **G-III** catalyst shows an extremely fast initiation and propagation rate which allows the preparation of polymer architectures with relatively low dispersities,[59] and as a result this catalyst has been widely employed as the prime catalyst by the ROMP community. Apart from these catalysts, ammonium tagged[60,61] and nitro-versions[62] of the **HG-II** catalyst, as well as metathesis catalysts based on Ru-cyclic alkyl amino carbene [Ru(CAAC)] complexes[63] have also been developed and commercialized.

5. Degradable Polymers by ROMP

ROMP can be applied to the synthesis of a wide variety of degradable polymers by the careful design of the cyclic olefin monomer. For the design of polymers with a degradable backbone by ROMP this requires a monomer with two important attributes: 1) it must be a strained cyclic or bicyclic olefin; and 2) it must contain core functionality which can impart a degradable linkage in the polymer backbone. Functional polymers with degradable backbones can also serve as scaffolds for directed drug delivery or regenerative medicine.[64,65].

In 1995, Grubbs *et al.* were pioneers in designing degradable cyclooctadiene/acetal copolymers by applying ROMP. The ROMP of cyclooctadiene (**M1**) with dioxepin monomers (type **M2**) were carried out in the presence of **G-I** to give copolymers having an acetal functionality in the polymer backbone (Scheme 3).[66] A copolymer of **M1** and 4,7–dihydro-2-phenyl-1,3-dioxepin (**M2b**) displayed degradation under acidic conditions (HCl, H_2SO_4, or trifluoroacetic acid) at the benzylidene acetal functionality to obtain 1,4-hydroxytelechelic polybutadiene (HTPBD). However, degradation of the copolymer with more stable



Scheme 3. ROMP of cyclooctadiene (**M1**) with dioxepin monomers (type **M2**) and their acid-catalyzed degradation.

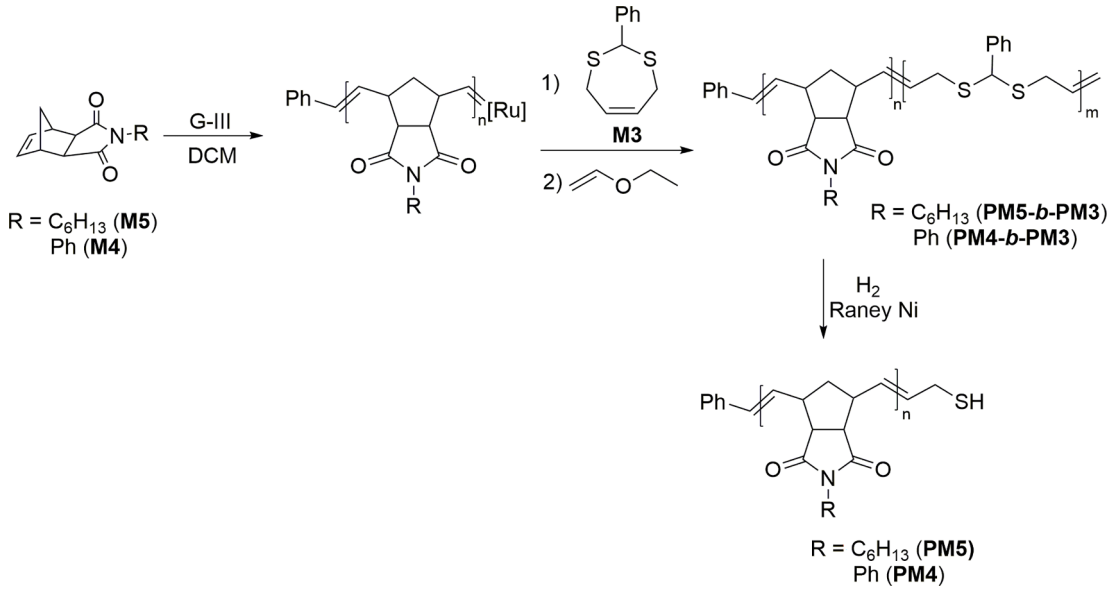
acetal linkages [**P(M1-*co*-M2a)**] required more forcing conditions.

Based on the above report, Kilbinger *et al.* introduced a novel approach for the synthesis of thiol end-functionalized ROMP polymers by the so-called 'sacrificial synthesis' route (Scheme 4). This route consisted of 1) the synthesis of diblock copolymers by sequential addition using living ROMP, and 2) subsequent treatment with conc. HCl (in $MeOH/CH_2Cl_2$) or with a reducing agent to cleave the olefinic thioacetal groups in the second block to obtain the end-functionalized polymers.[22,23,67] In a seminal experiment, diblock copolymers of exo-N-hexyl-2,3-norbornene dicarboximide (**M5**) or exo-*N*-phenyl-2,3-norbornene dicarboximide (**M4**) with unsaturated thioacetal monomer, 2-phenyl-1,3-dithiepine (**M3**), were synthesized in a **G-III** catalyzed ROMP.[67] The thioacetal block of the copolymers **P(M5-*b*-PM3)** and **P(M4-*b*-PM3)** were then cleaved by hydrogenation using Raney® nickel to access the thiol functionalized polymers **P(M5)** and **P(M4)** (M_n = 5.9–33.0 kg mol^{-1}, $Đ$ = 1.09–1.28) which were then employed for coating gold nanoparticles.

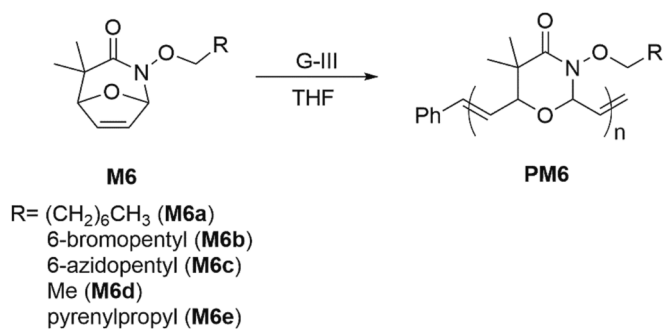
Fishman and Kiessling reported the synthesis and ROMP of bicyclic oxazinones monomers (type **M6**) to afford polymer scaffolds containing repeating oxazinone units within the polymer backbone (Scheme 5).[68] The synthesized polyoxazinones (type **PM6**) exhibited unique backbone-degradation profiles i.e. they showed degradability in either acidic or basic conditions, but were reported stable at neutral pH. For example, **P(M6e)** (M_n = 18.5 kg/mol) was subjected to degradation under a range of acidic and basic conditions at 25 °C. Analysis by size exclusion chromatography (SEC) indicated significant degradation of the polymer backbone at pH 2.5, with 66 % polymer mass loss within 6 h, while at pH values < 1 complete decomposition in under an hour was observed.

Degradable polyethylene analogues containing orthoester groups via ROMP were reported by Wurm and coworkers in 2019.[69] The ROMP of 1,5–cyclooctadiene (**M1**) with cyclic orthoester monomers (**M7**) in the presence of **HG-I** catalyst produced linear unsaturated polyorthoester-containing copolymers with M_n values up to 15 kg/mol ($Đ$ = 1.4–3.0) (Scheme 6). The acidic degradation of these polymers produced the corresponding alcohols and esters. The hydrolysis of **P(M7a-*co*-PM1)**, **P(M7b-*co*-PM1)**, and **P(M7d-*co*-PM1)** yielded methanol, while that of **P(M7c-*co*-PM1)** produced the less toxic isopropanol. Furthermore, the nature of the substituent at the orthoester group was demonstrated to influence the hydrolysis rate depending on the electron density induced by the R-group. The hydrolysis rate was reported to increase Me > H > CH_2Cl. These unsaturated polyorthoesters exhibited shelf lives of less than six months, indicating hydrolysis by atmospheric moisture. For example, the M_n of **P(M7c–*co*-PM1)** decreased from 8400 to 900 g/mol after storing the sample for six months without any precautions.

Later in 2020, Elling *et al.* reported the synthesis of degradable polyacetals and polyketals with controlled molecular weights (17.7–35.9 kg/mol) and low dispersities ($Đ$ = 1.07–1.25) by the application of alternating ring-opening metathesis polymerization (AROMP) of 1,1-disubstituted cyclopropenes (**M8**) and dioxepins (**M2b, M2d** and **M2e**) in the presence of **G-III** catalyst (Scheme 7).[70] Under the optimized



Scheme 4. Sacrificial synthesis of thiol-functionalized polynorborneneimides.

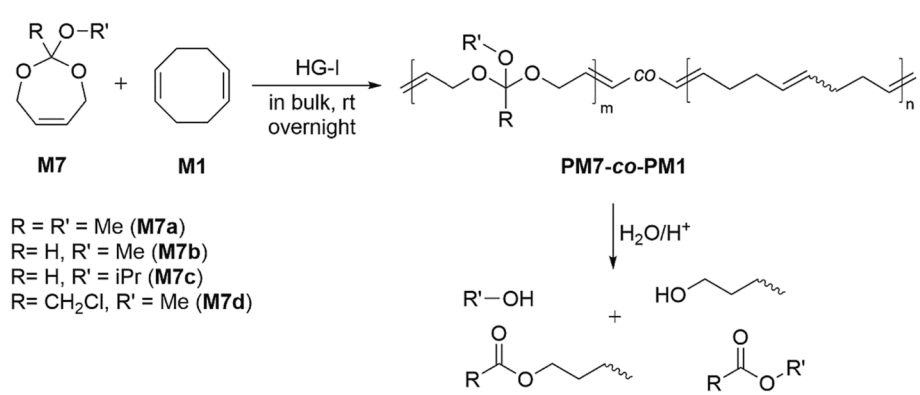


Scheme 5. ROMP of a bicyclic oxazinones monomer (type **M6**).

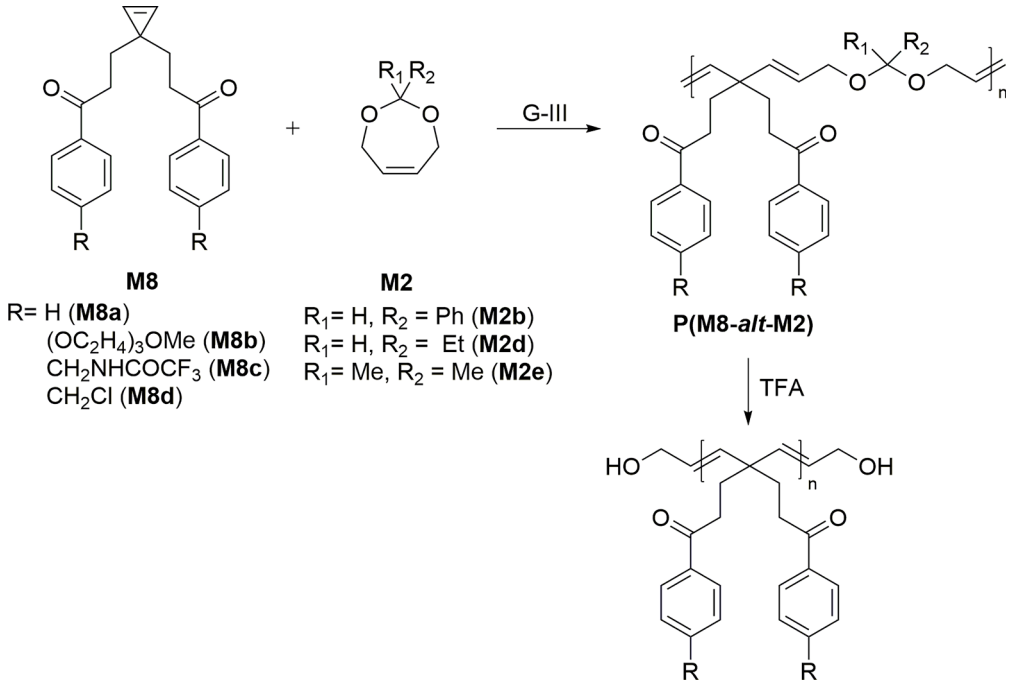
conditions of **M2**/**M8** = 1.25 at an initial concentration [**M8**] = 0.1 M, high degrees of alternation and controlled polymerization were achieved between non–propagating cyclopropenes and low-strain dioxepins. The high degree of alternation allowed the resulting polymers **P(M8-*alt*-M2)** to fully degrade under acidic conditions (1 vol% of TFA in THF) at variable rates dependent on the monomer structures. For example, the hydrolysis of the ketal–containing polymer, **P(M8a-*alt*-M2e)**, was observed to be the fastest, with complete degradation in less than 1 h due to the stabilized carbocation intermediate. However, the phenyl acetal polymer, **P(M8a-*alt*-M2b)**, degraded at a slower rate (<5h), while the ethyl acetal polymer, **P(M8a-*alt*-2d)**, was the slowest of the series, only degrading to about half of its original molecular weight after 9 h.

Inspired by these results, Sampson and coworkers applied the same AROMP strategy to incorporate acetal and ester functionalities into the backbone of copolymers, catalyzed by **G-III** catalyst.[71] Specifically, combining large-ring (7–10 membered), cyclic acetal (**M2**) or lactone (**M9**) monomers with monomer **M10** provided perfectly alternating copolymers **P(M10-*alt*-M2)** or **P(M10-*alt*-M9)** with acetal or ester functionality, respectively, in the polymer backbone (Scheme 8). Monomer **M10a** was found to be more reactive than **M10b**, theorized to be due to the shorter side chain decreasing steric hindrance during metathesis. Alternating copolymers **P(M10a-*alt*-M2a)** (M_n = 6.9–15.5 kg/mol, $Đ$ = 1.2–1.5) were also synthesized using equimolar ratios of **M10a** and **M2a** with catalysis by **G-III**. However, **P(M10a-*alt*-M2b)** copolymers (M_n = 7.5–13.2 kg/mol, $Đ$ = 1.2–1.5) were obtained with close to theoretical M_n values using a two-fold excess of monomer **M2b** with respect to **M10a**.

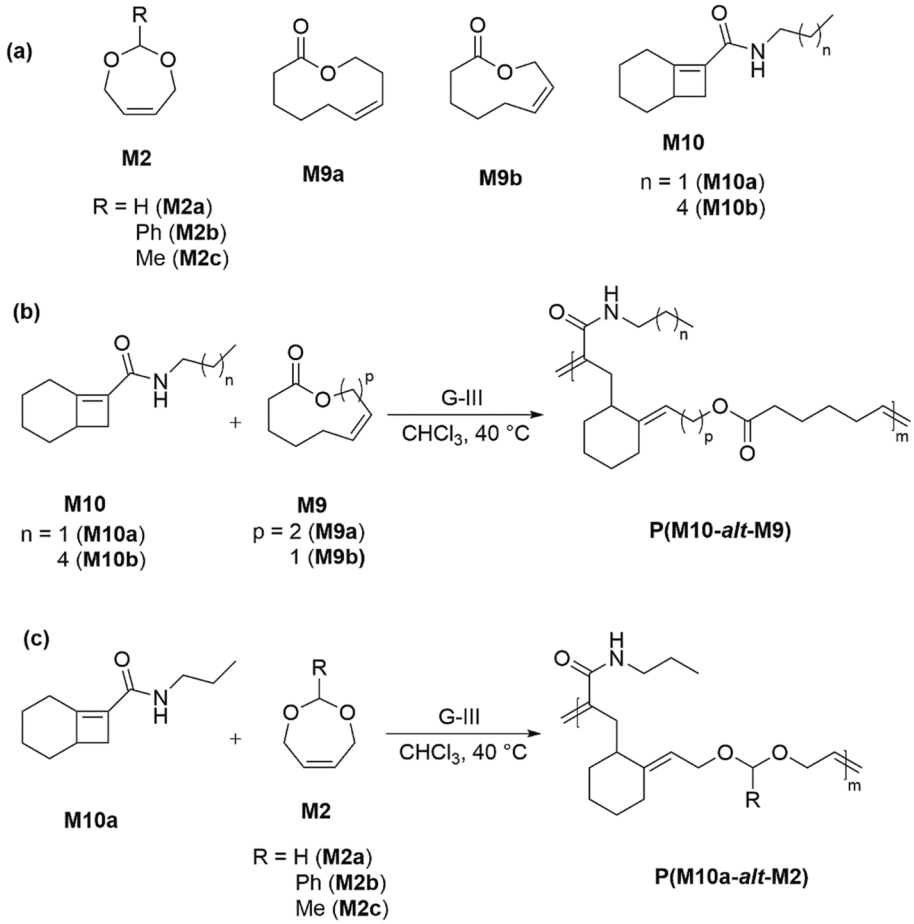
Due to the different solubility profiles of the ester- and acetal-containing copolymers, different hydrolysis protocols were applied for degradation. Ester-alternating copolymers were subjected to hydrolysis in aqueous solution at several pH values. Interestingly, **P(M10-*alt*-M9b)** was readily hydrolyzed under strong basic conditions; at pH 13, 78 % of **P(M10a-*alt*-M9b)$_{10}$** and 26 % of **P(M10b-*alt*-M9b)$_{10}$** were degraded within 8 h. However, under mildly acidic or basic conditions the **M9b-**



Scheme 6. Synthesis and hydrolysis of degradable polyorthoesters.



Scheme 7. Synthesis of degradable polyacetals and polyketals.



Scheme 8. (a) Structures of the monomers used for AROMP, (b) Synthesis of degradable **P(M10-*alt*-M9)** and, (c) **P(M10-*alt*-M2)**.

based copolymers exhibited no or slow degradation. For the degradation of acetal-containing copolymer **P(M10a-*alt*-M2)**, mild acidic hydrolysis conditions (sodium phosphate buffer, pH 5, at 37 °C) were applied. Under these conditions, **P(M10a-*alt*-M2b)$_{20}$** underwent complete hydrolysis within 30 h, forming degradation products corresponding to benzaldehyde and a diallyl alcohol derivative as confirmed by mass spectrometry (Scheme 9).

Sun, Gianneschi and colleagues reported the synthesis of a degradable phosphorus–containing polymer via ROMP using **G-III** catalyst.[72] Controlled ROMP of low ring-strain diazaphosphepine **M11** was achieved at low temperature to afford well–defined **P(M11)** that readily underwent degradation under acidic conditions via the cleavage of the acid-labile phosphoramidate linkages (Scheme 10a). For example, the ROMP of **M11** at 2 °C resulted in 55 % monomer conversion to obtain **P(M11)** (M_n = 14.2 kg/mol, $Đ$ = 1.16) in 5 h. To examine the degradation mechanism, both **M11** and **P(M11)** were subjected to 0.25 M HCl in DMSO-d_6. Over 48 h, the monomer **M11** completely degraded, while > 80 % of the polymer **P(M11)** degraded into phenylphosphoric acid within 240 h, as indicated by ^{31}P NMR spectroscopy. Monomer **M11** was also found to be compatible with the preparation of degradable random copolymers **P(M12$_{42}$-*co*-M11$_{32}$)** and **P(M13$_{38}$-*co*-M11$_{32}$)** ($Đ$ = 1.13 for both), and block copolymers **P(M12$_{50}$-*b*-M11$_{25}$)** ($Đ$ = 1.05) and **P(M13$_{40}$-*b*-M11$_{33}$)** ($Đ$ = 1.22) (Scheme 10b/c). Interestingly, both random and block copolymers, **P(M13$_{38}$-*co*-M11$_{32}$)** and **P(M13$_{40}$-*b*-M11$_{33}$)**, could self-assemble into spherical nanoparticles in aqueous solution. The cytotoxicity of the nanoparticles was evaluated and no appreciable cytotoxicity was observed even at concentrations as high as 150 μg/mL, indicating their potential as non-toxic nanocarriers.

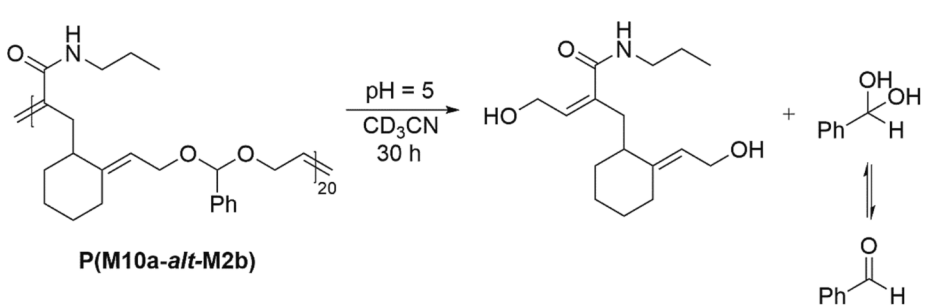
A similar approach was reported by Johnson and coworkers using silyl ether monomers (type **M14**) to prepare backbone-degradable statistical bottlebrush copolymers via ROMP.[73] A library of polymers of type **P(M14-*co*-PM15)** were synthesized using equimolar ratios of cyclic silyl ethers (**M14**) and norbornene analogue **M15** in the presence of **G-III** catalyst (Scheme 11). As analyzed by SEC, the copolymers displayed low dispersities ($Đ$ = 1.02–1.24) regardless of the silyl ether monomer used. Moreover, ^{1}H NMR analysis demonstrated high monomer conversion. Finally, acidic hydrolysis led to the degradation of the **P(M14$_{30}$–*co*-PM15$_{30}$)** copolymers, but no degradation of the homopolymer **P(M15$_{30}$)**. Very recently, the same group disclosed the synthesis of high-density polyethylene (HDPE)-like degradable polymers through a one-pot catalytic ROMP of **M14b**/**M14d** with cyclooctene followed by hydrogenation.[74].

Enol ethers have long been applied as quenching reagents for ROMP as they form a thermodynamically stable Fischer carbene with the Ru species. However, in 2021, Xia *et al.* reported a new kind of five-membered cyclic enol ether, 2, 3–dihydrofuran (**M16**), which was polymerized by an enthalpy-driven ROMP (in bulk using **G-I**/**G-II** catalysts) to yield degradable as well as depolymerizable poly (2, 3–dihydrofuran) **P(M16)** (Scheme 12).[75] High initial concentrations of **M16** promoted the polymerization reaction, with *ca.* 80 % monomer conversion after 4 h at 25 °C. The molecular weight of **P(M16)** was reported to vary linearly with catalyst loading, ranging from 6 to 127 kg/mol for a [M]/[I] value of 500/1 or 10200/1, respectively. The ether linkages contained within the polymer backbone of **P(M16)** also enabled degradation under acidic conditions.

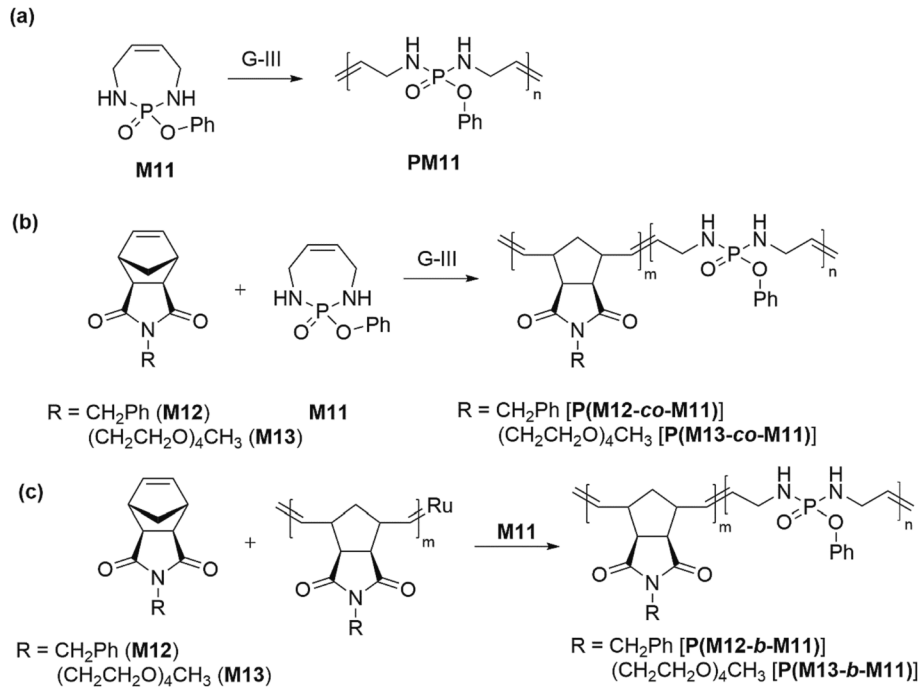
In 2022, Xia and Kilbinger independently reported the synthesis of degradable copolymers by the polymerization of 2,3-dihydrofuran (**M16**) with a variety of norbornene derivatives using **G-III** catalyst (Scheme 13).[76,77] It was demonstrated that the commercially available **M16** reacts rapidly with the **G-III** catalyst to form a thermodynamically stable Ru-Fischer carbene (the only detectable active Ru species during copolymerization), and that addition of the norbornene monomers becomes the rate determining step. This characteristic reactivity attenuates homo-addition of the norbornene monomer and therefore allows for uniform incorporation of acid–degradable enol ether linkages, enabling the complete degradation of the polymers under mild acidic conditions.

In a seminal experiment, the authors performed the ROMP of **M20** and **M16** using an equimolar feed (Scheme 14).[76] The resulting polymer, **P(M20-*co*-PM16)**, exhibited a $Đ$ of 1.03 with a well–controlled M_n of 19.3 kg/mol, close to the theoretical value (20.6 kg/mol), with the composition of **M20**:**M16** being *ca.* 100:80, respectively. **P(M20-*co*-PM16)** exhibited a T_g of 69 °C, higher than that of **P(M17-*co*-PM16)** (T_g = 21 °C) as a result of the more rigid norbornene imide structure. In addition, it was demonstrated that **P(M20-*co*-PM16)** was completely degradable upon hydrolysis with HCl in THF. However, living ROMP protocols require a stoichiometric amount of ruthenium catalyst with respect to the quantity of polymer chains. This leads to high catalyst loadings which is expensive, as well as difficult to purify due to the large quantity of toxic metal contaminants. It is therefore imperative that an alternative ROMP strategy is developed that applies catalytic amounts of ruthenium catalysts, making the process more environmentally conscious, as well as more useful, reliable and attractive for biomedical and technological applications.

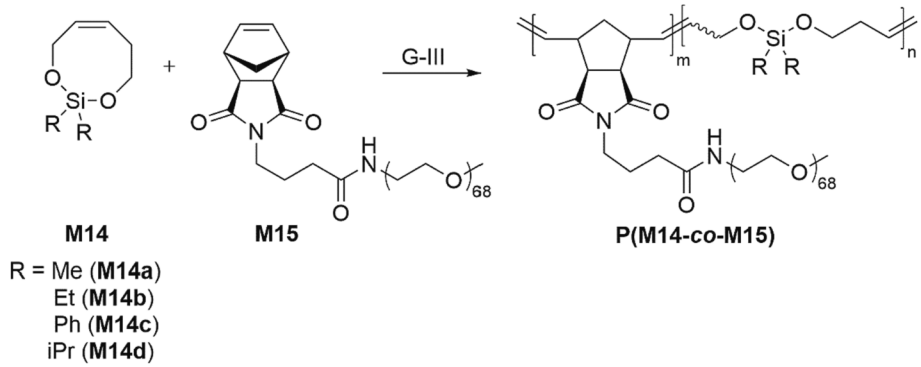
To address the above concerns, Kilbinger *et al.* reported a kinetically controlled catalytic ROMP (up to 500 times catalytic) by the application of vinyl ethers as chain transfer agents to synthesize readily degradable polymers with enol ether linkages via the copolymerization of **M16** with commercially available Vince lactam (**M24a**) and its derivatives (**M24b-d**).[77] A series of copolymers of type **P(M16-*co*-M24)** (M_n = 6.12–29.21 kg/mol, $Đ$ = 1.18–1.28) were synthesized by the living ROMP of **M16** and monomers of type **M24** (2:1 ratio, respectively) in the presence of **G-III** catalyst. A linear relationship was observed between the M_n values and the degree of polymerization, as expected for a living polymerization. To prove the living character of the copolymerization, a block copolymer was synthesized by the sequential polymerization of **M24c** and **M16**, followed by **M24b** and **M16** (Scheme 15). The SEC results displayed a complete shift of the first block (M_n = 2.23 kg/mol, $Đ$ = 1.14) to give the diblock copolymer **P(M16-*co*-M24c)-*b*-P(M16-*co*-M24b)** (M_n = 11.15 kg/mol, $Đ$ = 1.23). Addition of dilute HCl to a DCM solution of the copolymers gave complete degradation within 30 min due to the acid–labile vinyl ether linkages, which also suggested a uniform distribution of **M16** units throughout the backbone. Additionally, three reports for the synthesis of alternating degradable ROMP



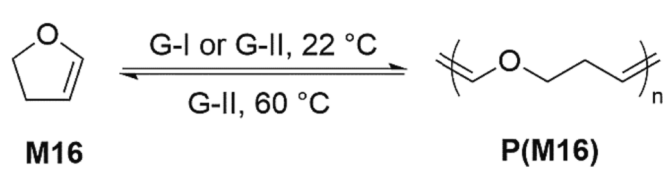
Scheme 9. Complete degradation of **P(M10a-*alt*-M2b)$_{20}$** within 30 h at pH 5.



Scheme 10. Synthesis of degradable polyphosphoramidates via ROMP of **M11**. (a) synthesis of the homopolymer of **M11;** (b) random copolymers and (c) block copolymers.



Scheme 11. Synthesis of statistical polymers of type **P(M14-*co*-PM15)**.



[M]/[I]	M_n (kg/mol)	Đ
500:1	6.0	2.14
1000:1	9.9	1.43
4000:1	47.4	1.60
10200:1	127.7	1.35

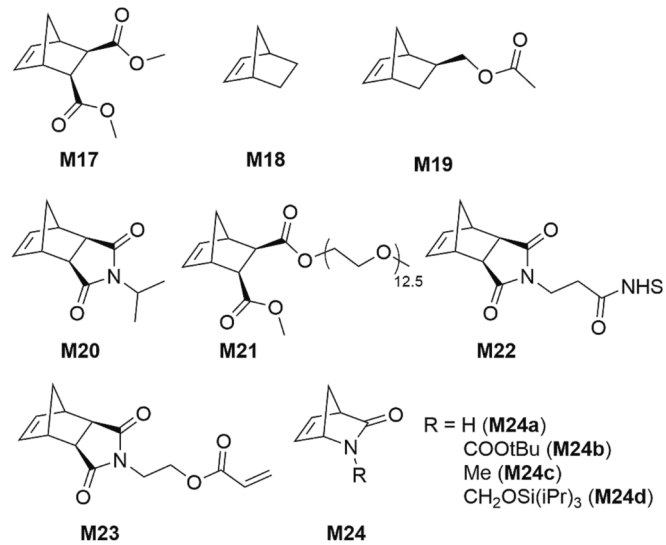
Scheme 12. ROMP of 2, 3-dihydrofuran (**M16**).

copolymers of **M16** with diversely functionalized *endo-/exo*-norbornenes and *endo*-tricyclo[4.2.2.0^{2,5}]deca-3,9-dienes have been published recently by the groups of Akiyama, Choi and Sun.[78–80].

6. Chemically Recyclable Polymers by ROMP

As mentioned in Section 3 (Mechanism of ROMP), all steps within ROMP are in principal reversible and therefore exist as an equilibrium between polymerization and depolymerization.[24] This characteristic property of ROMP has led to new avenues for the preparation of 'chemically recyclable' polymers. As chemically recyclable polymers can be depolymerized to give pristine quality monomer, the recovered monomer can be used to produce polymer articles of virgin quality, therefore establishing a genuine closed-loop lifecycle for plastics. As polymerizability in ROMP is generally governed by ring-strain energy (RSE), this can be tuned by varying the molecular structure of the cyclic olefin monomers.[81,82] The RSE value can also be selectively tuned to impart depolymerizability. The ideal monomer to afford chemically recyclable polymer is one which can efficiently polymerize at room temperature, and fully depolymerize upon a mild temperature stimulus.

Another factor that governs (de)polymerization is the ceiling temperature (T_c). Polymerization is favoured when the temperature of the system is below the T_c value, and conversely, depolymerization is favoured above the T_c value. Equally, when a polymerization system is held at the T_c the rate of polymerization and depolymerization is the same.



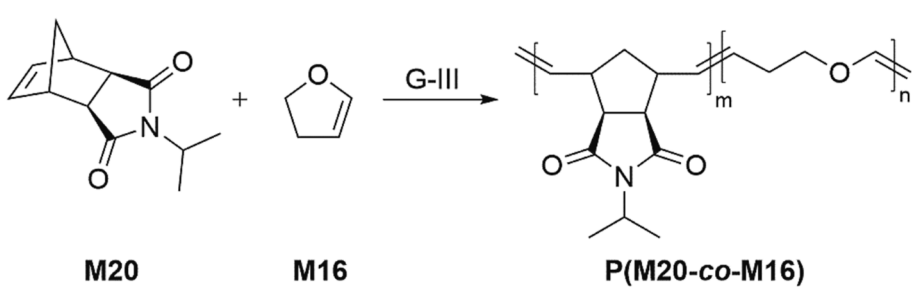
Scheme 13. Norbornene derivatives reported to undergo ROMP with 2,3–dihydrofuran (**M16**).

The T_c value can also be affected by the other reaction parameters such as monomer concentration, state and pressure of the system. An ideal chemically recyclable polymer system requires practical depolymerization conditions so that the depolymerization process is energy-efficient, while also not leading to monomer or polymer decomposition.[83] Thus, the balance between depolymerizability and thermal stability needs to be carefully considered.

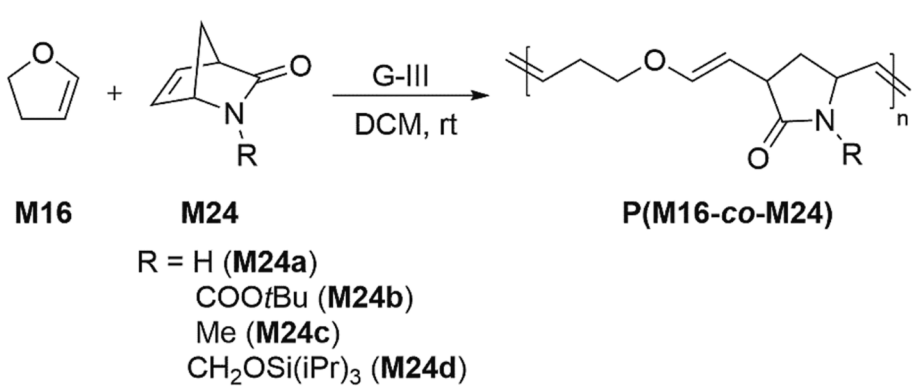
Since ROMP does not occur in the absence of a catalyst, unintended depolymerization can be prevented by simply removing or deactivating the catalyst, and thus ROMP can be considered a highly attractive tool for the synthesis of chemical recyclable polymers.[84] In addition, ROMP is also compatible with a wide range of functional groups and can be conducted under mild or even ambient conditions.[84] Moreover, compared to the other ring-opening polymerization (ROP) methods which are typically based on the ring-opening of cyclic esters or carbonates, ROMP enables the construction of polymers with hydrocarbon backbones, which have relatively greater hydrolytic and thermal stability.

Among the cycloalkene series, cyclopropene and cyclobutene are highly strained, with reported RSE values of 54.5 and 30.6 kcal/mol, respectively.[85] Due to their high RSE values, the depolymerization of their corresponding ROMP-derived polymers is not realistic. The RSE of cyclohexene is too low (2.5 kcal/mol) to achieve a successful ROMP (Scheme 16).[86] Cyclopentene and its derivatives have been applied for the preparation of depolymerizable polymers because of their intermediate RSE values (6.8 kcal/mol). In 1972, cyclopentene (**M25**) was reported to be (de)polymerizable by Ofstead and Calderon (Scheme 17).[87] Later, in 2005, Grubbs and coworkers also reported the fine balance between polymerization and depolymerization during the ROMP of cyclopentene.[86].

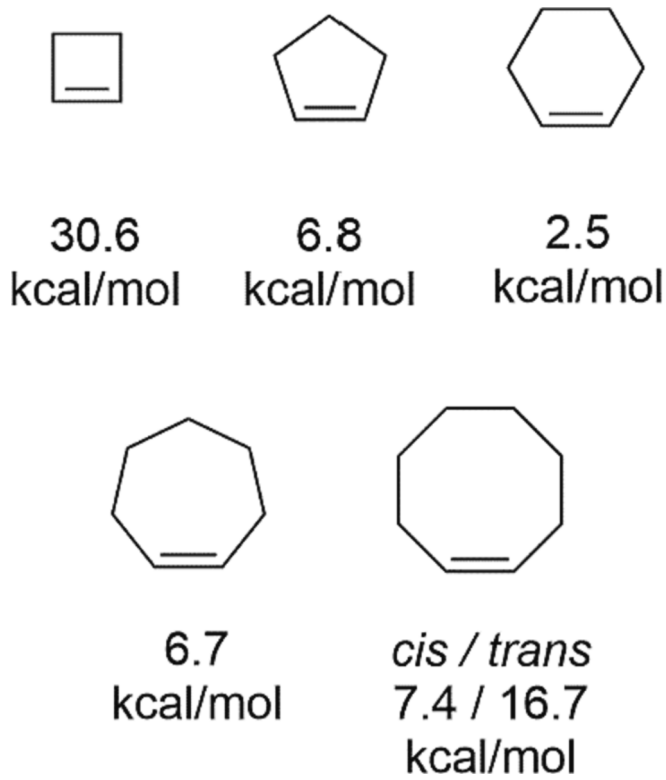
Bazzi *et al.* reported that cyclopentene (**M25**) and its silicon functionalized analogue **M26** could also be quantitatively (de)polymerized, proposing their use in recyclable tires.[88] **P(M25)** (M_w = 129 kg/mol, $Đ$ = 1.64) and **P(M26)** (M_w = 39.10 kg/mol, $Đ$ = 2.05) could be obtained by **G-II** catalyzed ROMP at ambient temperatures, and were readily depolymerizable to give quantitative monomer recovery by ^{1}H NMR (Scheme 17). It was reported that at high dilution, full depolymerization by **G–II** within 4 min at 25 °C could be achieved. Similar studies on the synthesis of **P(M26)** have been carried out by Kennemur *et al.* via variable temperature ROMP (VT-ROMP) of the highly strained **M26** monomer.[89] This transformation was achieved by careful consideration of the thermodynamics of low strain monomers, which allowed efficient initiation at warm temperatures followed by propagation to high conversion at low temperatures in a living-like fashion due to the suppression of chain transfer events. VT-ROMP of **M26** in the presence of THF and **G-I** proved optimal and narrow dispersities ($Đ$ < 1.2) were consistently obtained regardless of monomer-to-initiator ratio. Moreover, various cyclopentane analogues with T_c values ranging from 50 to 100 °C were reported by Moore and coworkers as useful candidates



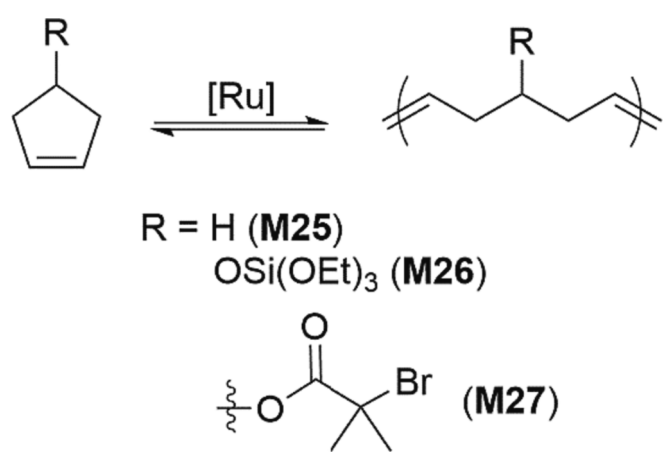
Scheme 14. Synthesis of degradable **P(M20-*co*-M16)** polymer.



Scheme 15. ROMP of 2,3-dihydrofuran (**M16**) with Vince lactam (**M24a**) and its derivatives (**M24b-d**).



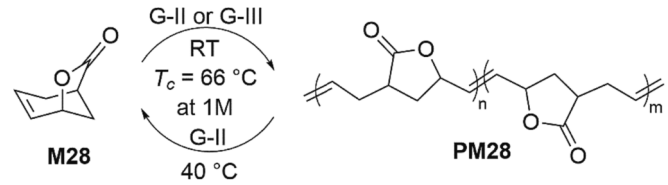
Scheme 16. Ring-strain energies (RSE) of cyclic olefins.



Scheme 17. (De)polymerization of polypentenamer monomers.

for the preparation of depolymerizable polymeric networks which can be reshaped and reformed.[90] Later, Kennemur and coworkers reported the synthesis of self-immolative bottlebrush polymers via the ROMP of cyclopentenyl monomer **M27** bearing an atom transfer radical polymerization (ATRP) initiator moiety (Scheme 17).[91] As shown in Scheme 12, Xia *et al.* also studied **G-II** initiated depolymerisation behaviour of poly(2, 3–dihydrofuran) **P(M16)**, which was obtained by an enthalpy-driven ROMP of **M16** (in bulk using **G-I**/**G-II** catalysts).[75] A sample of **P(M16)** containing residual catalyst demonstrated depolymerization when heated at 60 °C for 2.5 h, yielding *ca.* 90 % monomer recovery by distillation, without any side reaction.

Compared to the five-membered cycloalkenes, six-membered cyclic homologues (i.e., cyclohexene) are considered non-polymerizable because of their low RSE. In 2022, Chen *et al.* reported a bridged bicyclic cyclohexene monomer (**M28**) which could achieve ROMP (Scheme 18).[92] It was noted that the ROMP of **M28** proceeded rapidly at 25 °C

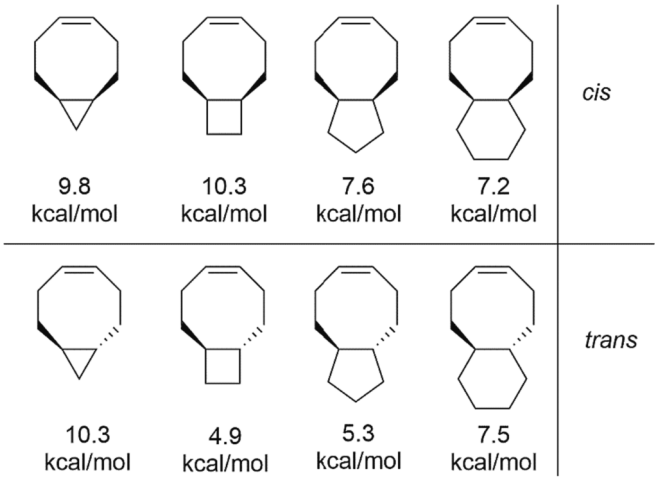


Scheme 18. Synthesis of recyclable polyolefin (**PM28**) from bridged bicyclic monomer (**M28**) based on cyclohexene.

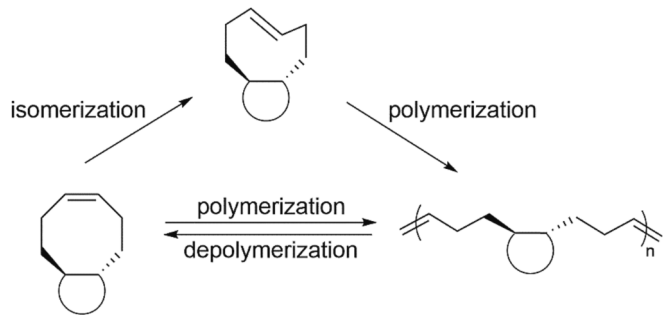
using **G-III** catalyst, however the measured M_n values were reported to deviate considerably from the calculated values. The application of **G-II** catalyst with catalyst loadings as low as 25 ppm provided better polymerization control, yielding **P(M28)** with M_n values as high as 340 kg/mol ($Đ = 1.30$). The depolymerization of **P(M28)** at 50 °C was reported to be facile in the presence of **G-II**, with successful depolymerization demonstrated on a 28 g scale to give 93 % monomer recovery. A similar report was also recently published regarding the synthesis of oxygen rich chemically recyclable polymers via the **G-III** catalyzed ROMP of *trans*-carbonate-fused cyclohexene monomers.[93] The deprotection of the carbonate group from the backbone of the polymers led to a well-defined 1,2-diol structure on every sixth carbon in the backbone, which could be recycled via closed- and open-loop recycling strategies to produce industrially useful chemical feedstocks.

Cyclooctene can be considered an attractive scaffold for use in ROMP as there exists convenient synthetic access to many analogues, but its high RSE (7.4 kcal/mol), as well as high T_c prevents its depolymerization. Computational DFT calculations demonstrated that the fusion of a cyclobutane or cyclopentane ring at the 5,6-position of cyclooctene could lower the RSE to 4.9 or 5.3 kcal/mol, respectively, thereby making these analogues attractive monomer choices for chemically recyclable polymers (Scheme 19).[94] Furthermore, it was noted that the RSE could be significantly increased through the isomerization of the *Z*-alkene isomer to the *E*-isomer (Scheme 20).

In 2021, Wang and coworkers reported significant advancements in the design of chemically recyclable polymers derived from ROMP by the (de)polymerization of diversely substituted cyclooctene analogues. A variety of cyclooctene monomers (**M29–M48**) with fused carbocyclic and heterocyclic rings at the 5, 6-position were synthesized, having comparatively lower RSE values (Scheme 21–22). Cyclooctene analogues containing fused cyclobutene rings (**M29–M34)** all underwent ROMP at [M] = 2 M in DCM using **G-II** catalyst to provide their respective polymers with over 80 % monomer conversions and M_n values of > 100 kg/mol.[83,94] It was noted that the stereochemistry and the nature of the substituent on the cyclobutene ring affect the T_g



Scheme 19. Ring-strain energies (RSE) of the fused cyclooctene series.

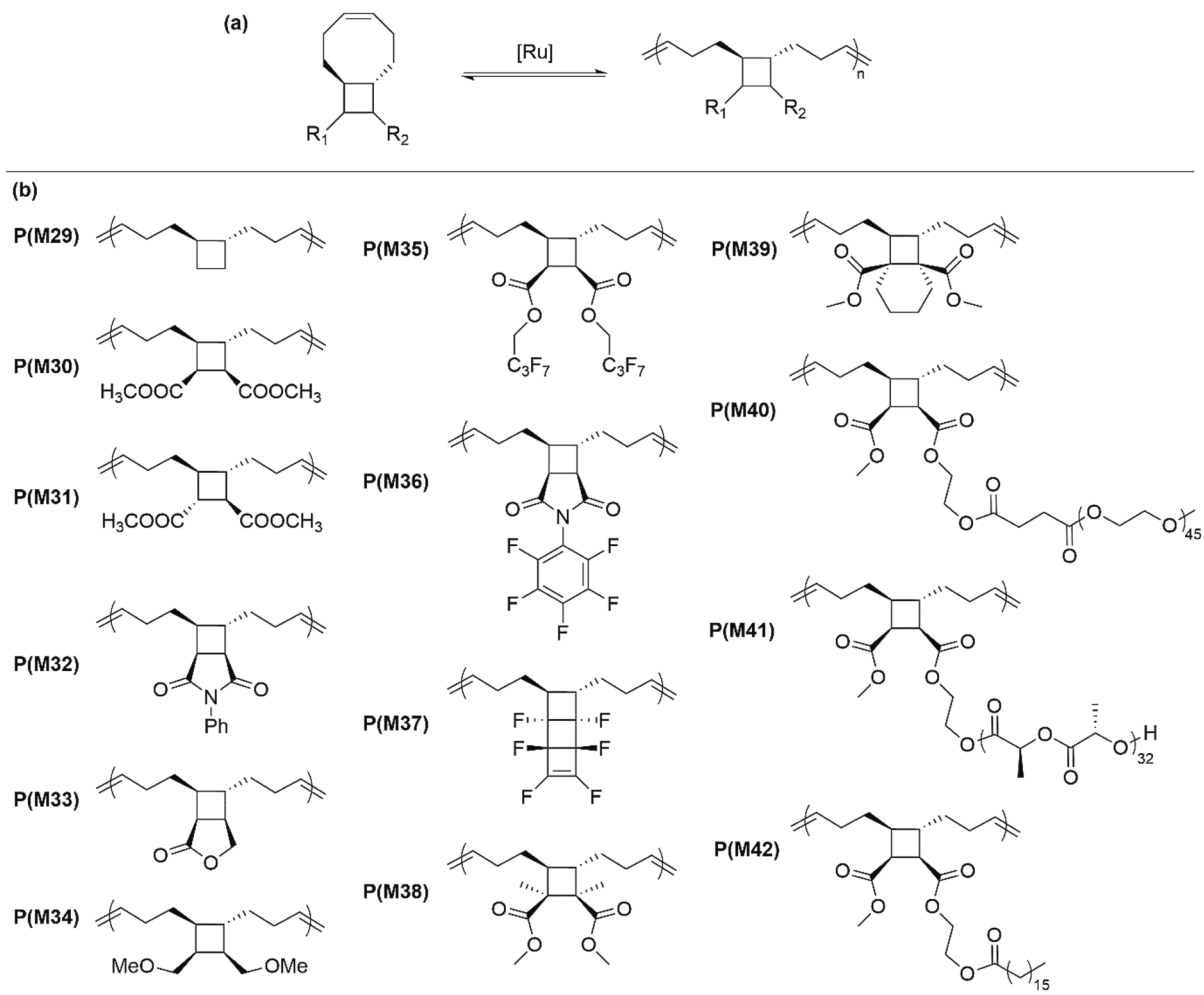


Scheme 20. Isomerisation, polymerization and depolymerization of fused cyclooctenes.

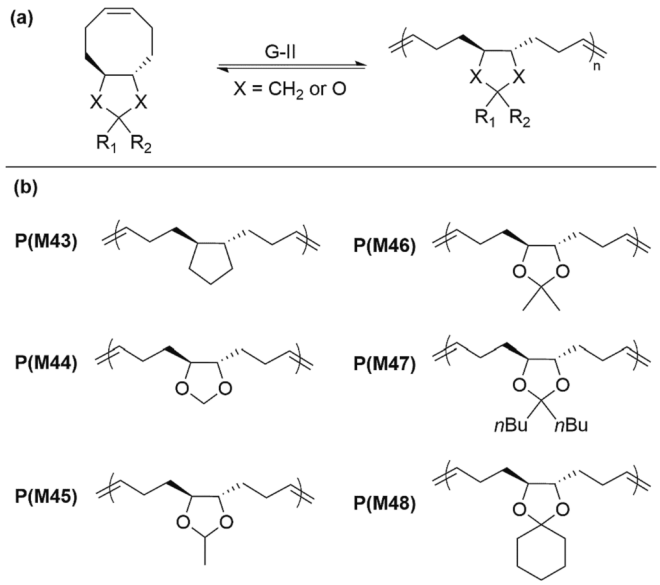
value and thus the material properties. For example, by the substitution of the protons of the cyclobutane ring in **M29** by methyl ethers (**M34**), methyl esters (**M30**), or a fused amide-containing ring (**M32**), the T_g value varies from −55 °C for the parent material to −34, 18, and 100 °C, respectively. Moreover, a considerable difference in T_g values i.e. 18 °C versus −1 °C, was observed for the respective *cis*-substituted **M30** and *trans*–substituted analogue **M31**.[95] All polymers in this series readily underwent depolymerization (>90 %) via ring-closing metathesis when heated with 1 mol% **G-II** in $CHCl_3$ at 50 °C. In addition, Wang *et al.* also reported monomers **M35–M37** to obtain fluorinated depolymerizable ROMP polymers with adjustable T_g values (2 °C to 88 °C), excellent thermal stability (initial decomposition temperatures > 280 °C) and hydrophobicity (water contact angles > 90°).[96] It can be concluded that the thermal and material properties of the cyclooctene-based polymers can be tuned by small structural changes in their respective monomer structures. In 2022, Wang *et al.* demonstrated a robust method for the synthesis of depolymerizable graft polymers **P(M40)–P(M42)** (M_n up to 14350 kg/mol and $Đ < 1.2$) derived from the controlled ROMP of their respective cyclobutane fused *trans*-cyclooctenes bearing side chains of poly(ethylene glycol), poly(L-lactide), and long alkyl chains (Scheme 21).[97].

Additional analogues of cyclooctene containing fused cyclopentane (**M43**) or cyclic acetal moieties (**M44**−**M48**) were also reported in an analysis of the structure property relationships of these polymers (Scheme 22).[83] Each monomer was subjected to ROMP using **G–II** catalyst under ambient conditions in DCM at an initial monomer concentration of 2.0 M and a monomer to catalyst ratio of 500:1 or 1000:1. Polymers with high molecular weights (M_n = 82–258 kg/mol) were obtained in all cases with dispersity values ranging from 1.6 to 2.1. The synthesized polymers displayed high thermal stabilities, with decomposition onset temperatures (T_d) above 320 °C, and T_g values ranging from − 55 to 8 °C (Table 1).

As displayed in Table 1, there was no significant difference in the T_c values between the cyclopentene fused analogue **P(M43)** (T_c = 614 °C) and its acetal counterpart **P(M44)** (T_c = 646 °C). Compared to **P(M44)**, the introduction of a single methyl substituent in **P(M45)** (T_c = 675 °C) did not significantly alter the T_c value, although the introduction of an additional geminal methyl group in **P(M46)** significantly reduced the T_c value to 376 °C due to the Thorpe-Ingold effect. A change of the geminal



Scheme 21. Structures of the polymers derived from the ROMP of cyclooctene monomers with fused cyclobutene rings.



Scheme 22. ROMP polymers of cyclooctene containing fused cyclopentane (**M43**)/cyclic acetal moiety (**M44** − **M48**).

Table 1
Reported thermodynamic and materials properties of the chemically recyclable polymers derived from fused cyclooctene monomers (**M29–M48**).

Polymer	Catalyst	M_n (kg/mol)	$Đ$	T_c (℃)	T_d (℃)	T_g (℃)
P(M29)	**G-II**	58	1.61	335	391	−55
P(M30)	**G-II**	142	1.55	199	401	18
P(M31)	**G-II**	81.8	2.10	–	361	−1
P(M32)	**G-II**	198	1.96	95	409	100
P(M33)	**G-II**	105	1.42	–	430	49
P(M34)	**G-II**	293	1.91	–	372	−34
P(M35)	**G-II**	276	1.42	–	335	−2
P(M36)	**G-II**	147	1.75	–	395	88
P(M37)	**G-II**	108.3	1.64	–	298	13
P(M38)	**G-II**	157	1.7		341	50
P(M39)	**G-II**	238	1.9		346	61
P(M40)	**G-I**	312	1.05	–	–	–
P(M41)	**G-I**	14,350	1.17	–	–	–
P(M42)	**G-I**	68	1.12	–	–	–
P(M43)	**G-II**	82	1.64	614	405	–33
P(M44)	**G-II**	171	1.71	646	360	−30
P(M45)	**G-II**	102	1.78	675	367	−14
P(M46)	**G-II**	247	1.89	376	342	−4
P(M47)	**G-II**	164	2.08	380	327	−27
P(M48)	**G-II**	258	1.86	571	347	−8

dimethyl groups to geminal di-*n*-butyl groups in **P(M47)** (T_c = 380 °C) did not significantly alter the T_c value. Interestingly, the introduction of a spiro-cyclohexane moiety made the T_c value rise to 571 °C for **P(M48)**. These various phenomena offer strong evidence that geminal disubstitution assists facile depolymerization.

7. Conclusion and outlook

The design and synthesis of degradable and chemically recyclable polymers not only offers an opportunity to recycle limited petroleum resources, but also to solve the environmental crisis caused by the accumulation of plastic waste. To achieve sustainability, the polymer industry has already taken the first steps in designing sustainable polymers, such as those created from renewable resources (i.e., polylactic acid). Nevertheless, although such plastics do display degradability under specific conditions, they possess poor thermomechanical properties. After multiple decades of development, a number of ROMP strategies have now also been developed for the preparation of sustainable polymers. At present, these strategies remain limited to the benchtop synthesis stage. Owing to the high costs and multistep monomer syntheses, it is presently difficult to envision their commercialization, especially as any new polymers must compete with the excellent material properties and low cost of synthesis of conventional vinyl-based polymers (i.e., polyethylene).

The preparation of the ROMP-derived polymers within this article remains energy intensive and often unselective, requiring harsh reaction conditions and hazardous chemicals for their production, degradation or depolymerization. It is thus imperative that new and efficient catalysts with high efficiency, selectivity, tolerance, and low cost are developed. It can be considered that the most important step in transitioning to a sustainable plastics economy is the design of the monomer itself. The molecular design of such degradable and depolymerizable polymers requires a judicious balance between material performance and recyclability. Most of the ROMP-derived polymers discussed within this article undergo degradation by acidic/basic hydrolysis, oxidation, reduction or photodegradation. It can be envisioned that these polymers also need to be evaluated for their biodegradation, by which they can naturally break down in the environment. Furthermore, depolymerizations using Ru catalysts are comparatively costly, to a similar extent as their synthesis. In general, efforts should be directed for the development of organocatalytic, cost-effective and/or non-toxic metal-initiated ROMP approaches.

The progression in degradable and chemically recyclable polymers using ROMP remains a developing area of research, and consequently there is a long path before their industrialization. It can be envisioned that these issues will be addressed in the near future, and that sustainable ROMP-derived polymers will have a significant impact on the plastics industry, as well as everyday life.

CRediT authorship contribution statement

Vishal B. Purohit: Writing – review & editing, Writing – original draft. **Marlena Pięta:** Writing – review & editing. **Joanna Pietrasik:** Writing – review & editing. **Christopher M. Plummer:** Writing – review & editing, Supervision, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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